18. Ans: A

$$V = \frac{Q}{4\pi\varepsilon_o r}$$

$$V \propto \frac{1}{r} \text{ (since Q is constant)}$$

$$\frac{V_P}{V_{surface}} = \frac{r_{surface}}{r_P}$$

$$V_P = \frac{0.10}{0.20} \times (-180)$$

$$= -90 \text{ kV}$$